

Post-Disaster Rescue Guidance via Fine-Tuning Vision Language Models on Aerial and Satellite Imagery

CSE499A/EEE499A/ETE499A – Senior Design I, Section 15 — Project Group 5

Abdullah Al Noman Tamanna Akter Mou Aryan Sami Ridita Afrin Riya Abrar Mohammed Tanzim Alam
ID: 2022095042 ID: 2211951042 ID: 2231407042 ID: 2211622042 ID: 2222864042

Abstract—Rapid, accurate identification of survivor locations following natural disasters remains a critical bottleneck in emergency response, with manual imagery interpretation creating throughput delays during the 72-hour golden rescue window. This project proposes a Vision Language Model (VLM) fine-tuning pipeline that processes post-disaster aerial and satellite imagery to generate structured, rescue-actionable textual guidance for response coordinators. We curate a multi-source instruction-following dataset from established disaster imagery repositories—xBD, FloodNet, AIDER, and RescueNet—formulating each training sample as an image-instruction-response triplet aligned to a standardised rescue guidance schema. We fine-tune Qwen2-VL-7B using parameter-efficient QLoRA adapters via Unsloth on Kaggle T4 GPUs, and benchmark systematically against GPT-4o Vision, LLaVA-1.5-7B, InternVL2-8B, and the untuned base model across ROUGE-L, BERTScore, and a novel Rescue Actionability Rubric (RAR). The fine-tuned model, dataset, and a Gradio application providing live inference with a benchmark comparison dashboard are released publicly on HuggingFace.

Index Terms—Vision Language Models; post-disaster response; satellite imagery; aerial imagery; QLoRA; instruction fine-tuning; rescue guidance; BERTScore; ROUGE-L; HuggingFace Spaces

I. INTRODUCTION

Natural disasters claim tens of thousands of lives annually, with a substantial proportion of casualties attributable not to the event itself but to delayed or misdirected rescue operations [1]. The first 72 hours after structural collapse or flooding, widely known as the *golden rescue window*, demands that coordinators make high-stakes deployment decisions rapidly and with incomplete ground truth.

Aerial and satellite imagery has become standard in disaster response, enabling wide-area damage assessment without exposing personnel to hazards [2]. Agencies such as FEMA, UN-OCHA, and the Copernicus Emergency Management Service routinely deploy remote sensing assets within hours of major events. While it is often assumed that aerial thermal cameras can bypass visual limitations to locate trapped survivors, rubble acts as a massive thermal insulator; materials like thick concrete completely block infrared radiation, and ambient heat from disaster environments frequently washes out heat signatures [12], [13]. Therefore, translating standard visual imagery into actionable rescue priorities, that is, identifying survivor zones, characterising structural collapse modes, and

flagging secondary hazards, is essential to guide ground operations. Currently, this relies almost entirely on trained specialists, creating a critical throughput bottleneck.

Recent Vision Language Models (VLMs), capable of jointly reasoning over images and generating coherent natural language, present a compelling opportunity to partially automate this interpretation task. Models such as GPT-4o Vision, Qwen2-VL [5], and InternVL2 [6] demonstrate strong visual reasoning across diverse domains. However, their ability to produce *rescue-actionable* language from disaster imagery, rather than generic scene descriptions, has not been rigorously characterised or domain-adapted. This project addresses that gap.

A. Problem Statement

Given a post-disaster aerial or satellite image, automatically generate a structured, rescue-actionable textual description that: (i) identifies priority zones for survivor extraction, (ii) characterises structural collapse mode and stability, and (iii) recommends areas requiring immediate avoidance. Because remote thermal imaging cannot reliably penetrate the insulating properties of dense structural rubble [12], this automated visual triage aims to reduce reliance on manual specialist interpretation, rapidly directing ground responders on exactly where to deploy time-consuming, penetrative physical sensors during time-critical operations.

B. Scope and Assumptions

In-scope: Single-image inference on aerial/satellite post-disaster imagery; VLM fine-tuning via QLoRA; instruction dataset curation from public repositories; multi-model benchmarking; Gradio deployment.

Out-of-scope: Real-time video analysis, multi-temporal change detection, autonomous rescue robotics, sensor fusion with ground-penetrating radar, and any surveillance or non-rescue application.

II. LITERATURE REVIEW

A. Post-Disaster Imagery Datasets

The xBD dataset [1] provides 22,068 pre/post-disaster satellite image pairs across 19 disaster types with per-building damage polygon annotations. FloodNet [2] supplies UAV

imagery post-Hurricane Harvey with pixel-level semantic labels. AIDER [3] provides aerial imagery across four disaster categories (fire, flood, collapsed building, traffic accident) while RescueNet [4] extends this with high-resolution drone imagery and rescue-oriented semantic annotations. Critically, none of these datasets include the instruction-following text responses required for VLM fine-tuning—a gap this project directly addresses.

B. Vision Language Models

LLaVA [7] established the VLM instruction fine-tuning paradigm by pairing CLIP visual encoders with LLaMA decoders and training on GPT-4 generated instruction data. Subsequent models have advanced visual reasoning through improved encoders and larger pretraining corpora. Parameter-efficient fine-tuning via LoRA [8] and QLoRA [9] has made domain adaptation feasible on constrained hardware, enabling 7B-parameter model fine-tuning on T4 GPUs.

C. AI for Disaster Response

Prior work has concentrated on damage classification [1], building footprint segmentation, and multi-temporal change detection. On the ground, Urban Search and Rescue (USAR) teams rely on highly specialised, micro-level penetrative tools, such as Delsar acoustic microphones, fiber-optic cameras, and radar, to confirm human presence, but these are time-consuming to deploy blindly across a disaster zone [13], [14]. The production of natural language rescue directives from aerial imagery, that is acting as a macro-level triage tool to direct where these micro-level physical sensors should be deployed, remains largely unaddressed, representing the primary novel contribution of this work.

D. Research Gaps

(i) No open-source VLM has been fine-tuned specifically for rescue-actionable text generation from disaster imagery. (ii) No standardised metric captures rescue relevance of generated text beyond surface-level lexical similarity. (iii) Systematic benchmarking of domain-fine-tuned open-source VLMs against GPT-4o Vision on disaster-specific tasks is absent from current literature.

III. OBJECTIVES AND KEY RESULTS

Objectives:

- O1. Curate a post-disaster VLM instruction dataset from multi-source aerial and satellite repositories.
- O2. Fine-tune Qwen2-VL-7B via QLoRA under resource-constrained hardware.
- O3. Establish a reproducible multi-model benchmark comparing fine-tuned, base, and SOTA VLMs.
- O4. Deploy the fine-tuned model with live inference and benchmark visualisation on HuggingFace Spaces.

Key Results (KRs):

- KR1. ≥ 500 verified instruction-response pairs across ≥ 3 disaster types and ≥ 2 data sources.

- KR2. $\geq 10\%$ ROUGE-L improvement over base Qwen2-VL-7B on held-out test split.
- KR3. Fine-tuned model achieves highest RAR score among all same-parameter open-source baselines.
- KR4. Publicly accessible Gradio application deployed on HuggingFace Spaces with live benchmark panel.

IV. PROPOSED METHODOLOGY

A. Dataset Curation

We aggregate imagery from xBD (satellite, multi-disaster), FloodNet (UAV, flood), AIDER (aerial, multi-disaster), and RescueNet (drone, rescue-oriented). Each sample is formatted as an instruction-following triplet:

```
{ 'image': 'post_quake_0042.jpg',
  'instruction': 'Analyse this aerial image and
  identify priority zones for rescue.',
  'response': 'Zone A (NE quadrant): pancake collapse, 3--4 floors. Extract at column intersections. Zone B (centre): lean-over, void likely on south face. Avoid SW -- full collapse, secondary risk high.' }
```

For images with existing annotations (e.g. xBD polygon labels), response text is auto-generated via GPT-4o Vision conditioned on annotation metadata, then manually reviewed and corrected against a standardised rescue guidance schema by team members. A 70/15/15 train/validation/test split is applied with no scene overlap across splits to prevent data leakage.

B. Model Architecture and Selection

We select Qwen2-VL-7B [5] as the primary fine-tuning target for its strong visual reasoning, native HuggingFace integration, and compatibility with Unsloth's optimised QLoRA implementation. The architecture pairs a Vision Transformer (ViT) encoder processing variable-resolution image tiles with a Qwen2-7B language decoder via a cross-modal adapter. A smaller Qwen2-VL-2B variant serves as the resource-constrained ablation target.

C. Fine-Tuning Strategy

We apply QLoRA [9] (4-bit NF4 quantisation; LoRA rank $r=16$, $\alpha=32$) to the language decoder attention projection matrices. Fine-tuning is conducted on Kaggle dual-T4 GPUs (2 $\times$ 16 GB VRAM) using Unsloth with memory-optimised gradient checkpointing and forced fp16 precision. The HuggingFace TRL SFTTrainer is used with a cosine learning rate schedule ($LR = 2 \times 10^{-4}$), effective batch size 16 via gradient accumulation, and early stopping on validation ROUGE-L.

D. Baselines

Table I. Benchmark Models

Model	Type	Params
GPT-4o Vision	Closed SOTA, API	—
Gemini 1.5 Flash	Closed, API	—
Qwen2-VL-7B (base)	Open, untuned	7B
LLaVA-1.5-7B	Open	7B
InternVL2-8B	Open	8B
Qwen2-VL-7B (ours)	Open, fine-tuned	7B

E. Evaluation Metrics

We evaluate on four dimensions. **ROUGE-L** [10] measures longest common subsequence overlap with reference responses. **BERTScore (F1)** [11] captures semantic similarity via contextual embeddings. The **Rescue Actionability Rubric (RAR)** is a novel human evaluation schema (1–5 Likert per dimension) scoring: zone identification specificity, collapse mode characterisation, non-expert actionability, absence of hazardous misguidance, and structural conciseness. **Zone Accuracy** measures correct priority zone identification against ground truth annotations where available.

The RAR constitutes the primary novel evaluation contribution of this work, motivated by the insufficiency of text-similarity metrics for assessing safety-critical guidance quality.

V. EXPERIMENTAL PLAN

Phase 1 (W1–2): Dataset collection, annotation schema design, GPT-4o response generation, and manual review for schema compliance.

Phase 2 (W3–4): Zero-shot inference of all baseline models on the held-out test set; metric harness setup; inter-annotator agreement measurement for RAR (κ target > 0.7).

Phase 3 (W5–7): QLoRA fine-tuning with hyperparameter sweeps (LoRA rank $\in \{8, 16, 32\}$; LR $\in \{1e-4, 2e-4\}$); ablation over data volume (25%, 50%, 100% of training set).

Phase 4 (W8–9): Full benchmark evaluation across all models; RAR human annotation (3 annotators, 50 samples each); comparison tables and error analysis.

Phase 5 (W10–11): Gradio deployment on HuggingFace Spaces; benchmark dashboard implementation; report writing and submission.

VI. RISKS AND MITIGATIONS

TABLE II
RISK REGISTER

Risk	Mitigation
Insufficient instruction pairs	GPT-4o annotation pipeline with manual review; targeted augmentation
T4 VRAM constraints	Qwen2-VL-2B fallback; gradient accumulation; Unsloth memory savings
BF16/fp16 CUDA conflict on T4	Force fp16 globally; verified Unsloth T4 compatibility patches
Hallucination in rescue output	RAR penalises hazardous misguidance; human review gate before deployment
Annotation subjectivity in RAR	Inter-annotator agreement check (κ); adjudication protocol

VII. ETHICAL CONSIDERATIONS

VLM outputs must never serve as authoritative rescue directives without expert human review; the deployed application includes a prominent disclaimer specifying the system as a decision-support tool only. Training data is audited for geographic and disaster-type representational bias, with failure modes documented explicitly in the model card. The system will not be trained on or evaluated against imagery depicting human casualties. Output logging is included in the deployment architecture to support post-hoc safety analysis.

VIII. DELIVERABLES

- Curated post-disaster VLM instruction dataset (HuggingFace Hub, public)
- Fine-tuned Qwen2-VL-7B QLoRA adapter weights (HuggingFace Hub)
- Benchmark evaluation and inference pipeline (GitHub, open source)
- Gradio application with live inference and benchmark dashboard (HuggingFace Spaces)
- IEEE-format 4-page final project report

IX. EXPECTED IMPACT AND LIMITATIONS

The fine-tuned model is expected to substantially outperform generic VLMs on rescue-specific language quality while remaining deployable on commodity hardware—lowering the barrier for integration into disaster response workflows in resource-limited settings. The open instruction dataset addresses a concrete gap in the disaster AI ecosystem and enables reproducible future benchmarking.

Limitations: Dependence on aerial image quality and sensor availability; potential failure on underrepresented disaster types or geographic regions; inherent language model hallucination risk in safety-critical contexts. The system explicitly does not replace structural engineering judgment or on-the-ground life detection. It acts solely as a decision-support triage tool to guide the deployment of human responders and penetrative physical sensors (e.g., sonar, radar, and gas detectors) [13].

X. CONCLUSION

This proposal presents a focused, feasible pipeline for generating rescue-actionable textual guidance from post-disaster imagery through VLM fine-tuning. By combining rigorous multi-source dataset curation, parameter-efficient adaptation under hardware constraints, a novel rescue-specific evaluation rubric, and systematic benchmarking against state-of-the-art models, the project targets a meaningful and concrete contribution to AI-assisted disaster response. Public deployment and open dataset release extend value beyond the course context, providing a reusable benchmark for the research community.

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